

# EDWIN LIM

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## EDUCATION

### National University of Singapore (NUS)

*Bachelor of Science in Data Science and Analytics*

- Minor in Computer Science
- Programmes & Societies: NUS FinTech Society, NUS Investment Society, NUS Science Club

Aug 2024 - May 2028

Singapore

## EXPERIENCE

### Tiktok

*Data Analyst Intern*

- Build and maintain dashboards to monitor external data labeller performance, enabling quality tracking across labelling operations.
- Conduct root cause analysis on data labelling quality issues and produce regular performance reports measured against established KPIs.
- Collaborate with Trust & Safety teams to translate data insights into business recommendations, supporting AI development initiatives.

Sep 2026 - Present

Singapore

### Monetary Authority of Singapore

*Data Analyst Intern*

- Conducted secondary analysis on survey datasets spanning 50+ companies and 20+ variables, applying feature engineering techniques such as column weighting to surface stronger signals, enabling the team to prioritize which industry factors warranted deeper focus.
- Developed a LLM-powered market intelligence platform tracking live Singapore news with on-demand company financial summaries.
- Fine-tuned AI agent prompts and conducted edge-case testing to improve system robustness and ensure comprehensive coverage.

Jan 2026 - Jun 2026

Singapore

### NUS Investment Society

*Quantitative Research Lead*

- Led a 4-person research project analyzing forecasted (FourCastNet) vs. realized (ERA5) weather data as directional signals for NATGAS futures, validating forecast skill (temperature MAE 0.58°C,  $R^2=0.94$ ) against realized weather.
- Engineered features from weather and futures datasets and developed a regime-switching model achieving ~82% directional accuracy.
- Developed a two-state regime-switching Hidden Markov Model with regime-specific logistic regression, estimated via the EM algorithm, achieving 75.0% test accuracy (a 16.7pp lift over baseline) at the 3-day horizon — the best-performing model in the study.

Aug 2025 - May 2026

Singapore

### NUS FinTech Society

*Machine Learning Analyst*

- Researched and implemented regime-switching (HMM) and EGARCH models for volatility forecasting, selecting model architecture and tuning parameters to balance sensitivity against false signals, as the primary contributor to the portfolio rebalancer's risk signal layer.
- Integrated regime and volatility signals as inputs to downstream portfolio rebalancing logic, interpreting model outputs to set actionable thresholds that enabled dynamic asset allocation under varying market conditions.

Jan 2026 - May 2026

Singapore

## PROJECTS & LEADERSHIP

### ML-Powered Smart Portfolio Rebalancer

- Researched and implemented Hidden Markov Model-based regime detection to identify latent high- and low-volatility market states.
- Applied EGARCH to capture asymmetric volatility clustering, supplying risk signals to a downstream portfolio rebalancing engine.

Jan 2026 - Jun 2026

### POLARIS: Policy-Aligned Local Review Inference System

- Built an NLP pipeline using Llama 3.2 to classify policy-aligned reviews, bootstrapping training data via ChatGPT-generated pseudo-labels due to lack of human-annotated ground truth, and achieving 87% agreement between the classifier and the reference labels.
- Applied pseudo-labeling via ChatGPT as a validation mechanism to bootstrap training data, improving model accuracy by 15%.

Aug 2025

### XGBoost Multi-Class Classifier for Corporate Structures | NUS Data-thon

- Developed an XGBoost multi-class classifier to identify corporate structures, achieving 77% accuracy, F1-score = 0.77, and AUC = 0.91.
- Engineered features from a dataset of 29,000+ companies across 22 variables, improving predictive power compared to baseline models. Collaborated in a team of 5 to conduct data cleaning, outlier handling, and model benchmarking (Logistic Regression, Random Forest).

Feb 2025

## SKILLS

**Languages:** English (Native), Chinese (Professional)

**Programming Languages:** Python, R, Java, SQL, MATLAB

**ML/DL Frameworks:** PyTorch, Scikit-learn, XGBoost, TensorFlow

**Data & Visualization:** Tableau, Streamlit, Matplotlib, Seaborn

**Tools:** Docker, Git, Microsoft Excel

**Interests:** Photography, Badminton, Retail Trading